could vary in a continuous manner, and there was no reason to treat it as if it were made up of grains. For example, the energy of a pendulum measures the amplitude of the swing. There seems to be no reason for a pendulum to oscillate only with certain determined amplitudes and not others. For Max Planck, taking energy in finite-size packets was only a strange trick which happened to work for the calculation – that is, to reproduce laboratory measurements – but for utterly unclear reasons.

Five years later it is Albert Einstein – him again – who comes to understand that Planck's packets of energy are in fact real. This is the subject of the third of the three articles sent to the *Annalen der Physik* in 1905. And this is the true date of birth of quantum theory.

In the article, Einstein argues that light truly is made up of small grains, particles of light. He considers a phenomenon that had been recently observed: the photoelectric effect. There are substances that generate a weak electric current when struck by light. That is to say, they emit electrons when light shines on them. Today we use them, for example, in the photoelectric cells which open doors when we approach them by detecting if light arrives, or not, in a sensor. That this happens is not strange, because light carries energy (it warms us, for example), and its energy makes the electrons 'jump out' of their atoms; it gives them a push.

But something is strange: it seems reasonable to expect that if the energy of light is scarce – namely, if the light is dim – the phenomenon would not take place; and that it would take place when the energy is sufficient – namely, when the light is bright. But it isn't like this: what is observed is that the phenomenon happens only if the *frequency* of light is high and does not happen if the *frequency* is low. That is to say, it happens or doesn't happen depending on the colour of light (the frequency) rather than its *intensity* (energy). There is no way of making sense of this with standard physics.

Einstein uses Planck's idea of the packets of energy, with a size that depends upon frequency, and realizes that if these packets are real, the phenomenon can be explained. It isn't difficult to understand why. Imagine that the light arrives in the form of grains of energy. An electron will be swept out of its atom if the individual grain hitting it has a great deal of energy. What matters is the energy of *each* grain, not the *number* of grains. If, as in Planck's hypothesis, the energy of each grain is determined by frequency, the phenomenon will occur only if frequency is sufficiently high, that is to say, if the *individual* grains of energy are sufficiently large, independently from the total amount of energy that's around.

It is like when it hails: what determines whether your car will be dented is not the total quantity of hail that falls but the size of the individual hailstones. There could be an enormous amount of hail, but it will do no damage if all the stones are small. In the same way, even if light is intense – which amounts to